

TOPIC 3.A

Structure, health, and diseases of the circulatory system

1. Describe the structure of the human circulatory system!
2. The health of modern humans is primarily threatened by diseases of the circulatory system. Which factors play a role in the proliferation of cardiovascular diseases?
3. What is called a myocardial infarction (heart attack)? What are its most important symptoms?
4. What causes atherosclerosis and what can be its consequences?
5. In your opinion, with what kind of lifestyle can diseases of the circulatory system be prevented?

Expected knowledge	Attainable score	Achieved score
1. • Humans have a closed circulatory system; vessels with muscular walls, arteries, transport blood from the heart, and they divide into capillaries. • From the capillaries, they gather into veins, transporting the blood towards the heart. • Structure of the heart. • Description of the systemic circulation. • Description of the pulmonary circulation.	1 point 1 point 5 points 2 points 2 points	
2. • Factors: stress, alcohol and nicotine consumption, overnutrition, and a sedentary lifestyle.	2 points	
3. • Infarction: if a coronary artery providing blood supply to the heart muscle becomes blocked somewhere, a part of the heart muscle dies due to a lack of oxygen. The blockage of a larger vessel and thus more extensive heart muscle death causes immediate death. • Symptoms: sudden pressure around the heart, chest pain, shortness of breath, fear of death, loss of consciousness. • Development: slowing down of circulation, calcification (hardening) of the vessel walls in the arteries.	2 points 2 points 2 points	
4. • Causes of atherosclerosis: - prolonged nervous system excitement (stress),	1 point	

- improper diet (consumption of fatty, cholesterol-rich foods), - smoking, - sedentary lifestyle. • Consequences: - the walls of the vessels become rigid, their diameter narrows, - the elastic elements of the vessel wall are destroyed, - high blood pressure develops - it primarily attacks the arteries of the systemic circulation.	2 points	
5. • The consumption of animal fats must be avoided; instead, plant-based oils should be preferred. • Overcoming situations that burden the mood and temperament. • Consumption of lots of vegetables, fruits, and low-fat dairy products. • Avoidance of smoking. • Regular exercise.	1 point 1 point 1 point 1 point 1 point	
6. Structure of the answer, communication skills	5 points	
Total:	30 points	

TOPIC 3.B

Characterisation of a plant species

1. Identify the received plant species using the Plant Identification book. Describe the process of identification.
2. To which major taxonomic groups does it belong? What are the characteristics of these groups that can be found on the received species?
3. Characterise the ecological requirements of the species.
4. In what type of habitat is the occurrence of this species typical? Where can we find such a place around the school?



Expected knowledge	Attainable score	Achieved score
1. Naming the species, listing the identification keys. (Common dandelion - <i>Taraxacum officinale</i>)	5 points	
2. Taxonomic group: phylum Angiosperms, class Dicotyledons • pinnate leaf venation, taproot system, • characteristic capitulum (flower head) inflorescence, • leaf blade is laterally lobed (pinnatifid), • fruit is a cypsela with a pappus.	5 points	
3. Ecological requirements: • moist, mesic areas, • nutrient-rich, slightly calcareous soil, • nitrogen indicator plant.	5 points	
4. Around the school, it occurs everywhere next to the fence in neglected areas. • Medicinal plant: appetite stimulant and cholagogue (stimulates bile flow), blood sugar-lowering, anti-atherosclerotic, anti-inflammatory (mention of at least 2).	5 points	
Total:	20 points	